

Assignment Report

Title: **Compiler Principles**

The Sixth Theory Homework

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[ABSTRACT]

After learning the lecture 6 of compiler principles[1], we got the homework at the end of the lecture.

There are some requirements for the homework:

- The file should be named as **“学号 _ 姓名 _ 第 i 次理论作业”**, where i is the number of the homework.
- The file should be in docx or **pdf** format.
- All the homework should be sent to the TA’s email: **406648896@qq.com**
- The deadline for the homework is **24:00 on Sunday of the week** when the homework is assigned.

Keywords: Compiler Principles

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1 Exercise 6.1

1.1 Problem

Consider the grammar:

$$S \rightarrow (L) \mid a$$

$$L \rightarrow L, S \mid S$$

We have the following operator-precedence relations for the grammar:

	a	(	)	,	$
a			>	>	>
(	<	<	≡	<	
)			>	>	>
,	<	<	>	>	
$	<	<			

Show the detailed process of the parsing of the sentence **(a, (a, a))**, follow the style in the previous slides.

1.2 Solution

Stack	Input	Action	Output
$	(a, (a, a)) $	Shift (	
$ (	a, (a, a)) $	Shift a	
$ (a	, (a, a)) $	Reduce	$S \rightarrow a$
$ (S	, (a, a)) $	Shift ,	
$ (S ,	(a, a)) $	Shift (	
$ (S , (	a, a)) $	Shift a	
$ (S , (a	, a)) $	Reduce	$S \rightarrow a$
$ (S , (S	, a)) $	Shift ,	
$ (S , (S ,	a)) $	Shift a	
$ (S , (S , a	)) $	Reduce	$S \rightarrow a$
$ (S , (S , S	)) $	Reduce	$L \rightarrow S, S$
$ (S , (L	)) $	Shift)	
$ (S , (L)	) $	Reduce	$S \rightarrow (L)$
$ (S , S	) $	Reduce	$L \rightarrow S, S$
$ (L	) $	Shift)	
$ (L)	$	Reduce	$S \rightarrow (L)$
$ S	$	Accept	

References

[1] JUN LI W. Lecture06. principles of compiler construction[Z]. 2026.